

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Third Semester of B. Tech. Theory Examination (CE / IT)****March -Apr 2018****MA201 Discrete Mathematics****Date: 26.03.2018, Monday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

- i. The question paper contains two sections.
- ii. Section I and II must be attempted in separate answer sheets.
- iii. Figures to the right indicate marks.
- iv. Make suitable assumptions and draw neat figure where it is required.
- v. All notations and terminologies are standard.

Section-I**Q-1 Select the correct option for following multiple choice questions. [07]**

- 1) Which of the following is not logically equivalent to $p \Rightarrow q$? **01**
 - a) $\neg q \Rightarrow \neg p$
 - b) $\neg p \vee q$
 - c) $\neg (p \wedge \neg q)$
 - d) $\neg p \Rightarrow \neg q$
- 2) Which one of the following is correct interpretation of given predicate $P(x) = \forall x : x \in \mathbb{Z} \Rightarrow (\exists y : (y \in \mathbb{R}) \wedge (y^2 = x))$? **01**
 - a) Every integer is square of some real number.
 - b) Square of every real number is some integer.
 - c) Every integer is some real number.
 - d) Every real number is some integer.
- 3) Let $(B, *, \oplus, ', 0, 1)$ be any Boolean algebra and $a, b \in B$. If $a \leq b$ then which of the following is not true? **01**
 - a) $a * b' = 0$
 - b) $a' \leq b'$
 - c) $a' \oplus b = 1$
 - d) $a' * b' = b'$
- 4) Every distributive lattice is also _____ lattice. **01**
 - a) Modular
 - b) Complemented
 - c) Complete
 - d) Bounded
- 5) Which of the following is not true? **01**
 - a) Every chain is a lattice.
 - b) Every POSET is a chain.
 - c) Every chain is a POSET.
 - d) Every lattice is a POSET.
- 6) A Boolean expression containing sum of minterms is known as _____. **01**
 - a) Symmetric expression
 - b) Product of sum canonical form
 - c) Sum of product canonical form
 - d) None of these
- 7) Which of the following is true for any Boolean algebra $(B, *, \oplus, ', 0, 1)$? **01**
 - a) $(B, *, \oplus)$ is a bounded lattice.
 - b) $(B, *, \oplus)$ is not a distributive lattice.
 - c) $(B, *, \oplus)$ is not a complemented lattice.
 - d) $(B, *, \oplus)$ is not a modular lattice.

Q-2 Attempt any four out of six . [16]

- 1) Check whether $\{\wedge, \neg\}$ is a minimal functionally complete set of connectives or not. 04
- 2) State and prove idempotent property for GLB and LUB of pair of elements of a lattice. 04
- 3) Consider the set of natural numbers $\mathbb{N}$ with the relation D - divides, then show that an algebraic structure $(\mathbb{N}, D)$ is a POSET. 04
- 4) Check whether the set $S = \{1, 5, 7, 35\}$ is a sub-lattice of (S_{385}, D) or not. 04
- 5) Let $(L, *, \oplus)$ be a lattice then prove $(a * b) \oplus (a * c) \leq a * (b \oplus c) \forall a, b, c \in L$. 04
- 6) Express the Boolean function $f(a, b, c) = ab + a'c$ as a product of max-terms. 04

Q-3

- 1) **Attempt any two out of three . [06]**
 - a) Check the validity of the given argument "If I drive to work then I will arrive in time. I do not drive to work. Therefore, I will not arrive in time." using inference rules of propositional logic. 03
 - b) Use truth table to obtain principal disjunctive normal form for the logical expression " $(\neg p \vee \neg p) \wedge (\neg p \wedge r)$ ". 03
 - c) In any Boolean algebra prove that $a = b \Leftrightarrow ab' + a'b = 0$. 03
- 2) Consider a Boolean algebra $(S_{91}, *, \oplus, ', 0, 1)$ with division relation, then find the value of a Boolean expression defined as $\alpha(x, y, z) = (x * y') \oplus (z' * y) \oplus (z * x')$ at a point $(1, 7, 91)$. [06]

OR

- 2) Show that $(S_{70}, *, \oplus, ', 0, 1)$ is a Boolean algebra with the division relation. [06]

Section-II

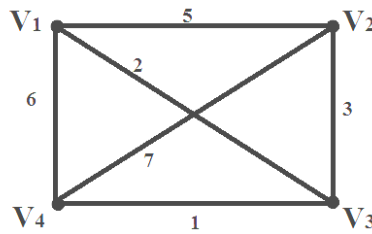
Q-4 Select the correct option for following multiple choice questions. [07]

- 1) A graph which contains exactly one edge between each pair of distinct vertices is _____.
a) Bipartite b) Connected c) Eulerian d) Complete 01
- 2) The maximum possible number of vertices in any k- level tree is _____.
a) $2^{k+1} - 1$ b) $2^k - 1$ c) 2^{k+1} d) $2^k - (k + 1)$ 01
- 3) The number of different adjacency matrices for a graph with 5 vertices is _____.
a) 5 b) 10 c) 24 d) 120 01
- 4) How many fundamental circuits are there in a graph with 6 vertices and 8 edges?
a) 5 b) 3 c) 2 d) 1 01
- 5) How many fundamental cut sets are there in a graph with n vertices?
a) n b) n - 2 c) n - 1 d) n! 01

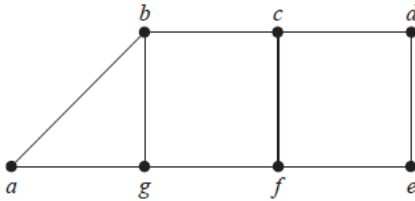
- 6) A connected planar graph with 10 vertices each of degree 3 split the plane into _____ regions. 01
 a) 7 b) 4 c) 2 d) 9
- 7) Which of the following is / are kuratowski's graph? 01
 a) K_5 b) $K_{3,3}$ c) both d) none

Q-5 Attempt any four out of six . **[16]**

- 1) Define the following terms: Tree, binary tree, minimal spanning tree and leaf. 04
- 2) Find three distinct Hamiltonian cycles of given graph. Also find their weights of each Hamiltonian cycle. 04



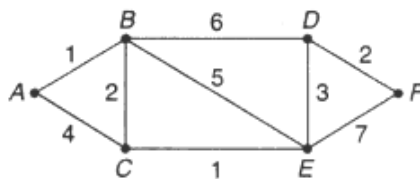
- 3) Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter? 04
- 4) Define Vertex connectivity and Edge connectivity. Find vertex connectivity and edge connectivity for the following given graph. 04



- 5) Define digraph and adjacency matrix. Also draw the digraph corresponding to adjacency matrix. 04

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

- 6) Find the shortest distance from A to F in following figure, using Dijkstra's Algorithm . 04



Q-6

1) Attempt any two out of three .

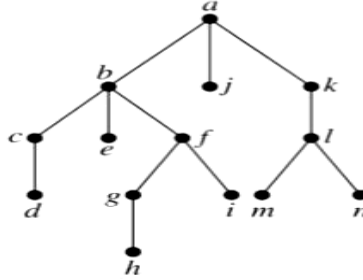
[06]

a) Draw the complete bipartite graphs of $K_{2,3}$ and $K_{3,4}$.

03

b) Consider the rooted tree T with root a given in following figure:

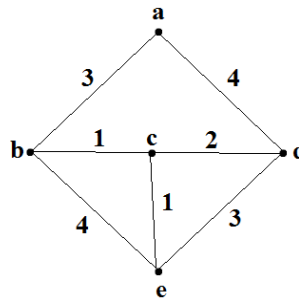
03



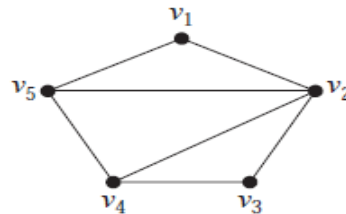
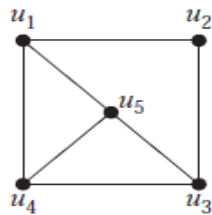
Find parent of f, ancestor of f, descendants of b, siblings of c, all leaves of tree and all internal vertices.

c) Find minimal spanning tree using Kruskal's algorithm for the given graph.

03



2) Define isomorphic graphs. Determine whether the given graphs are isomorphic or not. [06]



OR

2) Define a planar graph. Prove that $K_{3,3}$ and K_5 are non-planar graphs.

[06]